

Semantic network structure of an AI risk taxonomy reveals loss-of-control concepts as connector hubs

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ABSTRACT

Operational AI risk taxonomies are trees, although risk meanings overlap across category boundaries. We analyse the semantic space of a large taxonomy: 1,660 active level-4 risk cards distilled via a documented evidence chain from 3,906,767 research, policy, and patent records. Two expectation–maximization procedures structure the space: a constrained-EM audit stabilizing category assignment, and a graph-regularized seeded spherical EM partitioning the card network into 50 communities. In the released network, 27.7% of cards belong meaningfully to two or more level-3 families—the first quantification of what a single-parent tree discards. To measure connectivity as a property of the risks themselves, we rebuild the network as a similarity-threshold graph with no nearest-neighbour floor. Degree heterogeneity is substantial (Gini 0.50), yet formal tests reject a power-law tail. The semantic core is dominated by concepts of eroding human control and miscalibrated trust—automation bias, overreliance, delegation to misaligned agents, collectively inefficient multi-agent equilibria—which rank highest across thresholds, connect over 40 of 50 communities, and are far more likely to be under taxonomy review. This centrality is a structural property of system-level risk concepts, with implications for taxonomy design and review prioritization.

Introduction

Organizations that build or govern AI systems increasingly rely on structured risk taxonomies to inventory what can go wrong. Public efforts include harm taxonomies for language models¹, the MIT AI Risk Repository, which consolidates hundreds of risks from dozens of source frameworks², surveys of catastrophic AI risks³, and governance frameworks such as the NIST AI Risk Management Framework⁴ and the OECD classification of AI systems⁵. In operational practice these taxonomies share one structural commitment: they are trees. Every leaf-level risk is filed under exactly one parent category, because single-parent hierarchies are easy to audit, assign ownership over, and keep consistent across releases.

Risks, however, are not defined by their position in a tree. They are defined by natural-language descriptions of failure mechanisms, and the semantics of those descriptions overlap freely across category boundaries. A card about AI-generated scams is simultaneously about fraud, misinformation, and manipulation; a card about loss of human oversight touches nearly every operational category. A single-parent hierarchy must pick one shelf for each such concept and silently discard the rest of its meaning. How much is discarded, and which concepts sit where the discarding is most severe, has to our knowledge not been quantified for a production AI risk taxonomy.

Here we treat a large operational taxonomy—the KT RAI Risk Taxonomy 2.0, release v2.17.2—as an empirical semantic system and study it with the tools of network science⁶. The taxonomy’s 1,660 active level-4 (L4) risk cards descend, through a documented and reproducible evidence chain, from 3,906,767 research papers, policy reports, and patents. Each card carries a bilingual mechanism definition, and the released pipeline embeds these definitions⁷, links each card to its nearest semantic neighbours, and partitions the resulting proximity network into semantic communities.

This paper makes three contributions. First, we describe the two expectation–maximization (EM) procedures⁸ that give the taxonomy its semantic structure: a *constrained-EM audit* used during construction to test and stabilize category assignments under explicit guards, and a *graph-regularized seeded spherical EM*⁹ used to detect communities in the proximity network while respecting both embedding geometry and network topology. Second, we characterize the semantic proximity network itself and quantify cross-category overlap: 459 cards (27.7%) carry non-trivial responsibility in two or more level-3 (L3) families, which is precisely the information a single-parent tree cannot represent. Third, we characterize the connectivity of the risk space as a property of the risks themselves rather than of any particular graph construction: because the released union- k NN network imposes a degree floor and suppresses hubs, we recompute the card embeddings and rebuild the network as a similarity-threshold graph with no floor, subject the resulting degree distribution to formal tail diagnostics^{13,14}, and show that its semantic core is dominated by concepts of eroding human control and miscalibrated trust. We explain why such system-level concepts must occupy the centre of any sufficiently comprehensive AI risk space, connecting an empirical observation

about taxonomy structure to theoretical arguments about gradual human disempowerment^{10–12}.

Throughout, we report what the tail statistics actually support: the degree distribution is strongly heterogeneous but likelihood-ratio tests reject a power-law tail, so we make no scale-free claim^{13,14}. And we treat semantic proximity as a statement about meaning, not about causal transmission between risks¹⁵: the network records which failure descriptions resemble one another, not which failures cause one another.

Results

An evidence-grounded risk registry

The analysed release (v2.17.2) rests on a five-step distillation (Methods). An integrated AI evidence space of 3,906,767 records (1,749,159 papers, 125,372 policy reports, 2,032,236 patents) is filtered to a source-governed AI-risk evidence view of 82,971 records. Joining source risk identifiers and reviewing redundancy yields 1,726 candidate L4 rows; exact-label deduplication and removal of taxonomy-level leakage yield 1,711 canonical cards with permanent identifiers; retiring merged records leaves 1,660 active cards, each carrying a mechanism-level bilingual definition and its evidence references. All 1,660 active cards are assigned to one of the taxonomy’s L3 families, with 765 cards (46.1%) additionally carrying a HOLD marker indicating that the assignment is operationally usable but queued for further human review. Card-level scalar attributes such as estimated severity or probability exist in the registry but are intentionally excluded from all analyses reported here, because they are editorial estimates rather than measured quantities.

The semantic proximity network

Embedding each card’s bilingual mechanism definition with BGE-M3 and linking mutual nearest neighbours (base $k = 8$, minimum cosine similarity 0.62, projected union- k NN $k = 10$; edge weight mixes an L3-profile similarity with direct cosine similarity at 0.65/0.35; Methods) produces a network of 1,660 nodes and 11,869 edges. The network is a single connected component with density 0.0086, mean degree 14.30 (median 13), maximum degree 41, average clustering coefficient 0.333, and slightly positive degree assortativity (+0.093). Figure 1 shows the network in its released ForceAtlas2 layout¹⁷. This released network defines the taxonomy’s community structure and visual layout; because its union- k NN construction guarantees every node ten projected neighbours and truncates each node’s outgoing choices, we use a separate construction-free network, introduced below, for all degree and centrality analysis.

The graph-regularized seeded spherical EM partitions the network into 50 semantic communities (sizes 2–140, median 25), each seeded by an L3-family centroid but free to reorganize cards across family boundaries (Methods). The largest communities are Weaponization (140 cards), Policy Exposure (114), Non-Contestability (90), Anthropomorphism (71), and Misinformation/Disinformation (68). The EM converged in 16 iterations to a mean within-community cosine similarity of 0.751.

Quantifying what a tree discards

For every card we compute a responsibility distribution over L3 families from its neighbourhood structure (Methods). At a responsibility threshold of 0.03, 459 of 1,660 cards (27.7%) belong appreciably to two or more families, and individual cards reach up to eight family affiliations. In other words, more than a quarter of the taxonomy’s leaf concepts are multi-parent concepts being stored in a single-parent data structure. This is, to our knowledge, the first direct quantification of the representational loss imposed by tree-shaped AI risk taxonomies, and it identifies exactly which cards would benefit from an explicit multi-label extension.

Cross-community membership is the norm rather than the exception at the level of neighbourhoods as well: in the construction-free network introduced below, the median participation coefficient¹⁶ across all cards is 0.880, meaning that a typical card’s semantic neighbours are spread over many communities.

Degree structure without construction artifacts

Degree in the released network is not a clean property of the risks: the union- k NN construction imposes a hard floor at $k = 10$ and truncates every node’s outgoing choices, compressing heterogeneity (Gini 0.153) and suppressing hubs. To measure connectivity as a characteristic of the risk concepts themselves, we recompute the BGE-M3 embeddings of all 1,660 cards (reproduction fidelity against the released edge similarities: $r = 0.980$, mean absolute error 0.014; Methods) and build a similarity-threshold graph: two cards are linked if and only if their cosine similarity is at least θ , with no neighbour count imposed anywhere. At the pipeline’s own minimum-similarity value $\theta = 0.62$ this yields 90,848 edges, mean degree 109.5 (median 69), maximum degree 554, five isolates, a giant component of 1,655 nodes, clustering 0.432, and assortativity +0.197.

Freed from the floor, the degree distribution is strongly heterogeneous: the Gini coefficient more than triples to 0.504, the maximum degree is eight times the median, and 162 cards (9.8%) exceed four times the median (Fig. 2a). Heterogeneity is not an artifact of the specific threshold: across $\theta \in [0.55, 0.70]$ the Gini coefficient rises monotonically from 0.33 to 0.58 while

degree rankings remain stable (Spearman $\rho = 0.89$ between $\theta = 0.62$ and $\theta = 0.68$; $\rho = 0.94$ against the fully threshold-free strength $\sum_j s_{ij}$; Fig. 2b). Formal tail diagnostics, however, reject a power law: likelihood-ratio tests¹³ favour lognormal and exponential alternatives over a power-law tail (all $p < 0.001$), consistent with the observation that genuinely scale-free networks are rare¹⁴. The risk space is therefore best described as broadly—but not scale-freely—heterogeneous: a substantial semantic core of highly connected concepts over a large periphery of specific mechanisms.

Concepts of eroding human control dominate the semantic core

Table 1 lists the ten highest-degree cards of the construction-free network. The single most-connected concept in the entire risk space is RAI4-0592, “Collectively inefficient equilibria among AI agents” ($k = 554$), and its rank is robust: it is first at every threshold from 0.60 to 0.65 and second at 0.68. More broadly, the top of the ranking interleaves two kinds of concepts: system-level erosion of human control and trust—automation bias (rank 2), delegation of decision-making power to misaligned AIs, misaligned rogue behaviour, overreliance undermining user autonomy, anthropomorphic overtrust, miscalibrated trust, loss of human agency and autonomy—and broad misuse and integrity concepts (abuse and misuse, systemic bias, misinformation dissemination, insufficient security). Coding the 30 highest-degree cards by mechanism, 16 of 30 describe the erosion of human control, autonomy, or calibrated trust over AI systems. The loss-of-control theme is thus not carried by a single labelled card but saturates the semantic core, and it sharpens as the threshold tightens: at $\theta = 0.68$, overreliance (RAI4-1552) ranks first and loss of human agency (RAI4-0894) fifth.

Transparency requires noting what changed relative to the released construction: RAI4-1616 (“Future AI systems might actively reduce human control”), the highest-degree node of the union- k NN network, ranks 58th (top 3.5%) in the construction-free graph (Fig. 2d)—its former first place partly reflected mutual-selection dynamics of the k NN projection rather than raw semantic reach. The thematic conclusion survives the correction; the specific figurehead does not, which is precisely why the construction-free analysis matters.

These hubs are connectors, not local centres. The four highest-ranked control-erosion cards have participation coefficients¹⁶ of 0.92–0.94 and neighbourhoods spanning 42–47 of the 50 communities; top-decile hubs generally sit at the 87th–100th percentiles of P . In the role terminology of Guimerà and Amaral, the semantic core of the taxonomy is built from connector hubs: concepts whose neighbours are distributed across many communities rather than concentrated in their own.

HOLD status rises monotonically with degree: 63.5% of cards in the top degree decile carry HOLD markers versus 44.1% of the remainder, against an overall rate of 46.1% (Fig. 2c). We interpret this not as a defect of the hub cards but as a structural signal: cards whose meaning spans many families are precisely the cards for which a single released parent is hardest to defend, and the review process detects this.

Table 1. The ten highest-degree L4 risk cards in the construction-free similarity-threshold network ($\theta = 0.62$, release v2.17.2 card set). k : degree; s : strength (summed similarity weight); community: EM community of the card in the released network; P : participation coefficient (percentile in parentheses); review: HOLD marks cards queued for taxonomy review. Cards marked † describe erosion of human control, autonomy, or calibrated trust.

ID	Label (abbreviated)	k	s	Community	P	review
RAI4-0592†	Collectively inefficient equilibria among AI agents	554	366	Conflict	0.94 (96)	HOLD
RAI4-1482†	Automation bias	527	349	Non-Contestability	0.94 (97)	—
RAI4-1013	Abuse and misuse	523	347	Weaponization	0.93 (89)	HOLD
RAI4-1566	Systemic bias across specific communities	499	328	Hate and Unfairness	0.94 (96)	—
RAI4-0844	Disseminating false or misleading information	494	327	Misinf./Disinf.	0.94 (99)	HOLD
RAI4-1186	Insufficient security measures	489	319	Weaponization	0.93 (87)	—
RAI4-1038	Misapplication of AI	487	321	Goal Misalignment	0.95 (99)	—
RAI4-1425†	Delegating decision-making power to misaligned AIs	487	322	Goal Misalignment	0.93 (92)	—
RAI4-0565†	Misaligned rogue AI behavior	485	318	Goal Misalignment	0.95 (100)	—
RAI4-1719	Lack of security and resilience	485	319	Non-Contestability	0.93 (93)	HOLD

Discussion

Why loss-of-control concepts must sit at the centre

The empirical finding is that concepts of declining human control occupy the top of the degree ranking and act as cross-community connectors. We offer three mutually reinforcing explanations.

First, *semantic generality*. Control-erosion cards describe system-level failure conditions—human oversight weakening, agency shifting to machines, trust miscalibrated—rather than domain-specific mechanisms. Almost every concrete risk de-

scription (a manipulated consumer, an unaccountable decision, an autonomous weapon, an addictive companion) contains, as part of its meaning, a local instance of humans losing control over an outcome. In embedding space, the general concept therefore sits within the similarity threshold of many specific ones, and degree records exactly that reach. The same mechanism places broad misuse concepts high in the ranking, which is why generality alone does not fully explain the pattern: among comparably general concepts, those about control and trust are systematically over-represented (16 of the top 30).

Second, *bridging position*. The hubs' participation coefficients of 0.92–0.95, with neighbourhoods spanning more than 40 of the 50 communities, show that their connectivity is not concentrated in one region: these cards lie on short semantic paths between operational communities (manipulation, governance, safety, economic concentration). A reader moving conceptually from “persuasive design” to “concentration of decision power” passes naturally through “humans no longer steer the system”. Connector-hub status is the network signature of that conceptual throughway.

Third, *consilience with theory*. Independent theoretical work argues that the most structurally important AI risks are not exotic single failures but gradual erosions of human control: the assistance-game framing of Russell¹⁰, the argument by Bengio and colleagues that advanced agentic systems require management of catastrophic and loss-of-control scenarios¹¹, and the gradual-disempowerment analysis of Kulveit and colleagues¹². The specific hubs we find read like that literature's mechanism list: automation bias, overreliance, delegation to misaligned agents, and collectively inefficient multi-agent equilibria are precisely the incremental channels through which gradual disempowerment is argued to proceed. Our result is an empirical, bottom-up counterpart: when nearly seventeen hundred risk concepts distilled from four decades of literature, policy, and patents are laid out by meaning alone, the concepts these theories place at the centre of concern also sit at the centre of the semantic space. The taxonomy was not designed to produce this: community seeds, guards, and audits operate at the family level, and no step of the pipeline privileges any card.

Implications for taxonomy design and review

Two practical consequences follow. First, the 27.7% multi-affiliation rate argues for multi-label extensions of operational risk taxonomies: a released single parent can remain the operational assignment while secondary family memberships are stored alongside, exactly where our responsibility distributions are non-trivial. Second, degree and participation supply a principled review-prioritization signal. The monotonic rise of HOLD status with degree—63.5% in the top decile—indicates that the human review process already gravitates toward semantically ambiguous, high-centrality cards; making that signal explicit would let review effort track structural importance directly.

Limitations

Three caveats bound the interpretation. Semantic proximity is not causal transmission: our network records similarity of failure descriptions, not pathways along which one failure triggers another, and coupling of the causal kind requires separate evidence of mechanism and temporal order¹⁵. Absolute degree values still depend on the choice of similarity threshold; our claims therefore rest on rankings and concentration measures shown to be stable across thresholds ($\rho \geq 0.89$) and against the fully threshold-free strength centrality ($\rho = 0.94$), not on any single graph. Degree in a similarity graph also rewards semantic generality as such—broad catch-all labels rank high alongside substantively central concepts—which is why we report the thematic composition of the core rather than treating any individual rank as meaningful by itself. Finally, the underlying registry, while evidence-grounded and bilingual, reflects the coverage of its source corpora; risks under-represented in research, policy, and patent literature are under-represented here as well.

Methods

Evidence chain and registry

The integrated AI evidence space contains 3,906,767 records: 1,749,159 papers (global Web of Science AI collections), 125,372 policy reports (curated AI-policy master plus policy-materials collection), and 2,032,236 patents (AI patent master). Records are standardized to a common schema, retained only with a non-empty title, and deduplicated within document type and analytical category by an ordered key hierarchy (patent application ID, DOI, source record ID, URL, normalized title–year–institution fallback). A source-governed AI-risk evidence view selects 82,971 records (50,375 papers, 938 policy reports, 31,658 patents). Joining source risk identifiers through a crosswalk and reviewing semantic redundancy yields 1,726 candidate L4 rows; exact normalized-label deduplication (semantic-similarity thresholds are deliberately not used) and removal of fourteen taxonomy-level leakage rows yield 1,711 canonical cards with permanent identifiers RAI4-0001 through RAI4-1726; registry-preserving consolidation of duplicate lineages leaves 1,660 active cards, with all retirements kept as traceable `merged_into` records. Card-level severity, probability, and impact attributes are excluded from every analysis in this paper.

Algorithm 1: constrained-EM audit

During taxonomy construction, category assignments are tested with a constrained expectation–maximization audit rather than unconstrained nearest-centroid reassignment. Each audit allows only a selected set of source-family cards to move, treats all other cards as fixed anchors, excludes protected destinations (the 182-card Physical AI subtree), and alternates between (E) scoring each movable card against candidate family centroids and (M) recomputing those centroids, until no assignment changes. The composite score of card i against family k is

$$S(i, k) = 0.60c(i, k) + 0.30d(i, k) + 0.10w(i, k), \quad (1)$$

where c is cosine similarity to the current family centroid, d is cosine similarity to the family definition, and w is TF-IDF keyword cosine similarity. A move is accepted only if the destination improves on the source by at least 0.020 and has keyword cosine of at least 0.015, with additional destination-specific definition guards. Audited moves remain HOLD until human adjudication. An unguarded control run on revised definitions produced 151 candidate moves, used as sensitivity evidence only; the guarded release records 22 defensible proposals, of which 14 change the operational parent.

Semantic proximity network

Each active card’s bilingual mechanism text (English label and definition, with Korean as supplementary signal) is embedded with BGE-M3⁷ (pinned snapshot) and ℓ_2 -normalized. A base mutual- k NN graph is built with $k = 8$ and minimum cosine similarity 0.62; a projected union- k NN graph with $k = 10$ defines the released edge set, so every node holds at least its own ten projected neighbours (degree floor). Edge weights combine an L3-profile similarity (cosine similarity between the cards’ family-responsibility profiles) and the direct embedding cosine at 0.65/0.35. The released two-dimensional layout is ForceAtlas2¹⁷ with fixed seed 42 and is used for display only.

Algorithm 2: graph-regularized seeded spherical EM

Communities are detected with a spherical (von Mises–Fisher-style) EM on the unit-normalized embeddings⁹, seeded by L3-family centroids (seed weight 5) and regularized by the network: the E-step responsibility of community g for card i combines an embedding term $\exp\{\beta \mathbf{x}_i^\top \boldsymbol{\mu}_g\}$ ($\beta = 18$) with a graph term weighting the community memberships of i ’s network neighbours ($\lambda = 2$), and the M-step re-estimates unit-norm community centroids from responsibility-weighted cards. The procedure converged in 16 iterations (final change rate 0, final mean within-community cosine similarity 0.751), yielding 50 communities of sizes 2–140. Multi-affiliation counts use the final responsibility distributions with threshold 0.03.

Construction-free similarity network

For degree and centrality analysis we rebuild the network without any nearest-neighbour machinery. The bilingual card texts (English label and definition, followed by the Korean label and definition) are re-embedded with BGE-M3 (dense embeddings, FlagEmbedding implementation, ℓ_2 -normalized) and validated against the released artefact: across the 11,869 released edges, the recomputed cosine similarities match the stored direct similarities with $r = 0.980$ and mean absolute error 0.014. The construction-free graph links cards i and j if and only if $\cos(\mathbf{x}_i, \mathbf{x}_j) \geq \theta$, weighted by the cosine. The primary threshold $\theta = 0.62$ is the released pipeline’s own minimum-similarity parameter; sensitivity is reported for $\theta \in \{0.55, 0.58, 0.60, 0.62, 0.65, 0.68, 0.70\}$, together with the fully threshold-free strength centrality $s_i = \sum_j \cos(\mathbf{x}_i, \mathbf{x}_j)$. Rank stability across these choices is summarized with Spearman correlations.

Network statistics and tail diagnostics

Statistics are computed with NetworkX: degree, strength, clustering, degree assortativity, Gini coefficients, community size distributions, and participation coefficients $P_i = 1 - \sum_g (k_{ig}/k_i)^2$ over the 50 EM communities¹⁶ of the released network. Percentiles for P are computed over all 1,660 cards. Tail diagnostics follow Clauset and colleagues¹³ as implemented in the `powerlaw` package: a discrete power law is fitted above an estimated $x_{\min}$ (maximum-likelihood $\hat{\alpha} = 1.93$, $x_{\min} = 42$) and compared with lognormal, exponential, stretched-exponential, and truncated power-law alternatives via normalized likelihood-ratio tests; the power law is rejected against every alternative (all $p < 0.001$). Thematic coding of the 30 highest-degree cards as control-erosion versus other was performed by the author from the card definitions and is reproducible from the released registry. Analysis code that reproduces every number and figure in this paper is available with the data (Data availability).

Use of artificial intelligence tools

Claude (Anthropic) was used to assist with analysis scripting, figure preparation, and manuscript editing under the author’s direction. The author reviewed and takes full responsibility for all content.

Data availability

The complete release data for taxonomy v2.17.2, including the semantic proximity network (`semantic_proximity_network.json`), card coordinates and registry (`risk_space.json`), and an interactive explorer, are publicly available at <https://github.com/deep1003/RAI-Risk-Taxonomy-2.0>. A versioned archive will be deposited on Zenodo with a DOI upon acceptance.

Code availability

The analysis scripts that reproduce all statistics and figures (`embed_cards.py`, `threshold_network.py`, `network_stats.py`, `make_figures_v2.py`) are available in the same repository under the MIT licence.

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Author contributions

Y.C. designed the study, conducted all analyses, prepared the figures, and wrote the manuscript.

Competing interests

The author is employed by KT Corporation, which develops and maintains the RAI Risk Taxonomy analysed in this work. The author declares no other competing interests.

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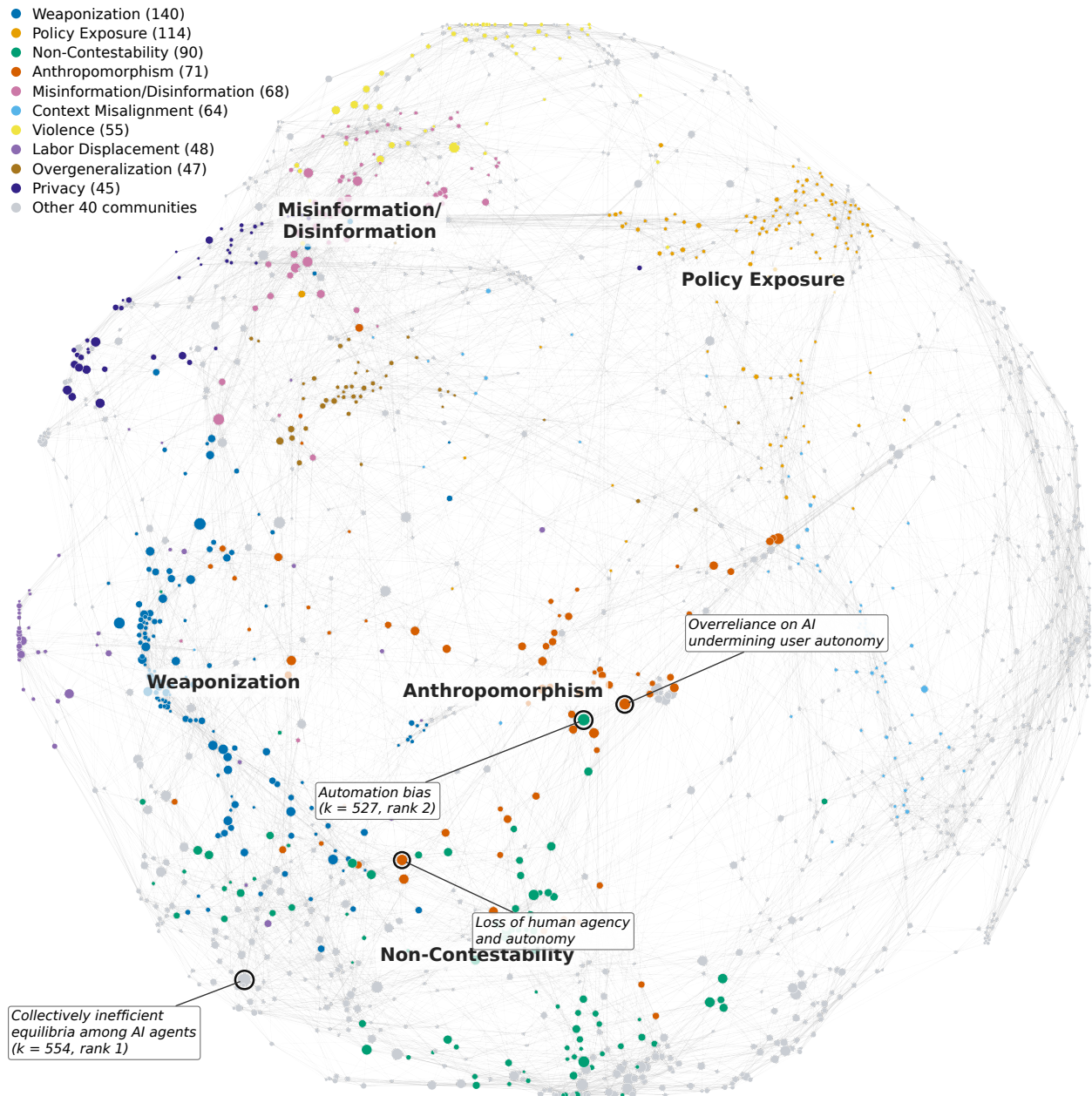


Figure 1. The semantic proximity network of 1,660 active L4 risk cards. Nodes are risk cards positioned by the released ForceAtlas2 layout; node area scales with construction-free degree (similarity-threshold graph, $\theta = 0.62$); colours mark the ten largest of the 50 EM-detected semantic communities (legend, with sizes), and grey marks the remaining 40. Edges of the released union- k NN network (11,869) are drawn in light grey. Black rings and italic labels mark four of the highest-ranked control-erosion hubs (RAI4-0592, RAI4-1482, RAI4-1552, RAI4-0894), which sit in different communities and regions of the space, consistent with their role as cross-community connectors. Bold labels mark the centroids of the five largest communities. The layout is used for display only; no analysis in this paper depends on layout coordinates.

